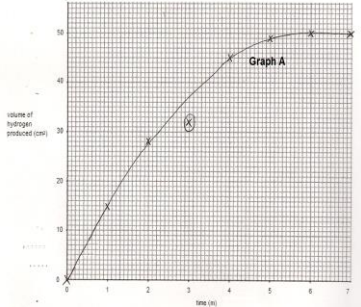
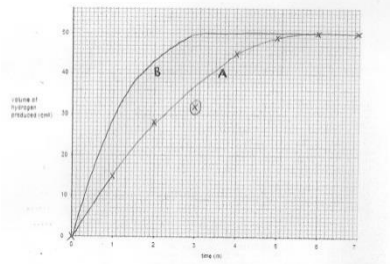


Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			fizzing / bubbling / effervescence (1) magnesium used up / gets smaller / disappears (1) neutral answers: exothermic / gas forms / hydrogen forms	2			2		2
	(b)			$\text{Mg} + 2\text{HCl} \longrightarrow \text{MgCl}_2 + \text{H}_2$ correct reactants (1) correct products (1) balancing (1) – can only be awarded if both the reactants and products are correct		3		3	1	
	(c)	(i)		 all plots correct (2) tolerance $\pm \frac{1}{2}$ square any 5 plots correct (1) curve missing out point at 3 minutes (1) no credit for joining points using a ruler		2		1	3	3

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)	I	21 / 22 cm ³ (accept without unit) accept alternative answers correctly read from the graph	1			1	1	1
			II	6 minutes / min(s) unit needed accept correct answer in seconds / minutes and seconds accept alternative values between 5 and 6 minutes based on graph		1		1	1	1
		(iii)		 steeper gradient (1) reaching same end point in less time (1)			2	2		2
				Question 4 total	3	6	3	12	6	6